

Technical University of Crete

Telecommunication Systems I

Exercise 1

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Hours required to complete the exercise: 3 hours

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1 Part A: Baseband communication with 2-PAM and truncated SRRC pulses

1.1 A.1 Constructing and plotting SRRC pulses

1.1.1 Statement of the question

Construct SRRC pulses $\phi(t)$ with $T = 10^{-2}$ s, $\text{over} = 10$, $T_s = T/\text{over}$, $A = 4$, and roll-off factor $a \in \{0, 0.5, 1\}$. Plot them on a common figure and state observations about the rate of decay of the pulse amplitude as absolute time t increases.

1.1.2 Figure

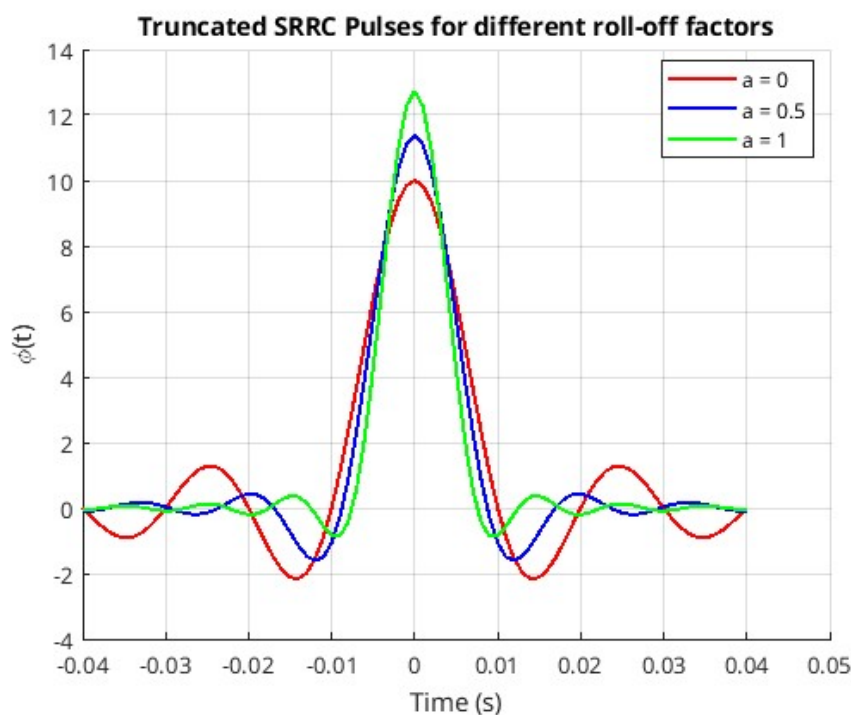


Figure 1: Truncated SRRC pulses for $a = 0$, $a = 0.5$, and $a = 1$.

1.1.3 MATLAB Code

```
1 %% A.1 Construct and Plot SRRC pulses
2 figure; hold on; grid on;
3 colors = ['r', 'b', 'g'];
4 for i = 1:length(a_values)
5     a = a_values(i);
6     [phi, t] = srrc_pulse(T, over, A, a);
7     plot(t, phi, 'Color', colors(i), 'LineWidth', 1.5, ...
8         'DisplayName', ['a = ', num2str(a)]);
9 end
10 xlabel('Time (s)'); ylabel('\phi(t)');
11 legend('Location', 'best'); hold off;
```

1.1.4 Comments or explanation

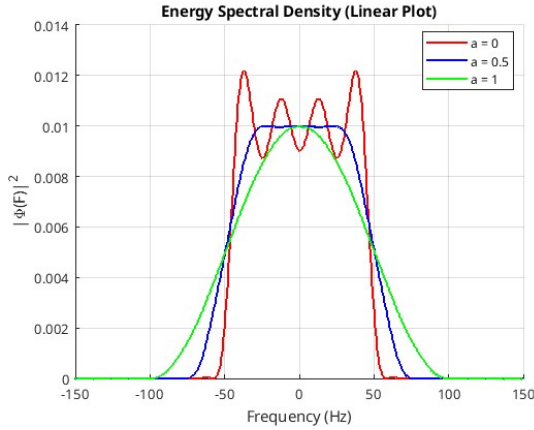
As the absolute time $|t|$ increases, the amplitude of the pulse decays towards zero. The rate of this decay is highly dependent on the roll-off factor a . The pulse with $a = 1$ decays the fastest, presenting very little ringing in its tails, while $a = 0$ decays the slowest. This occurs because a sharper transition in the frequency domain results in more spread in the time domain.

1.2 A.2 Fourier Transforms and Energy Spectral Density

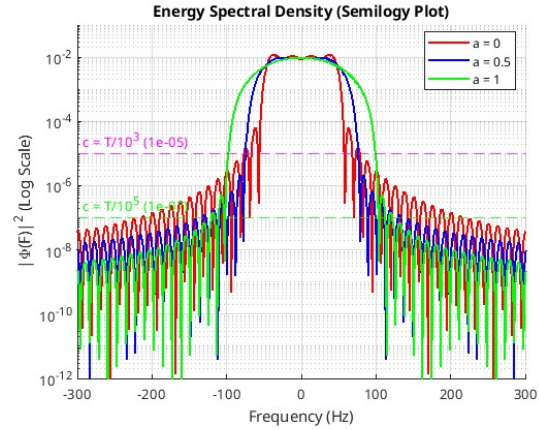
1.2.1 Statement of the question

Compute the Fourier transforms $\Phi(F)$ of the pulses at $N_f = 2048$ points over $[-F_s/2, F_s/2)$. Plot the energy spectral density $|\Phi(F)|^2$ on a common linear plot and a common semilogy plot.

1.2.2 Figures



(a) Linear Plot



(b) Semilogy Plot

Figure 2: Energy Spectral Density of SRRC pulses.

1.2.3 MATLAB Code

```
1 %% A.2 Fourier Transforms and Energy Spectral Density
2 N_f = 2048; Fs = 1/Ts;
3 F = linspace(-Fs/2, Fs/2 - Fs/N_f, N_f);
4 figure; hold on; grid on;
5 for i = 1:length(a_values)
6     a = a_values(i);
7     [phi, ~] = srrc_pulse(T, over, A, a);
8     PHI = fftshift(fft(phi, N_f)) * Ts;
9     ESD = abs(PHI).^2;
10    semilogy(F, ESD, 'Color', colors(i), 'LineWidth', 1.5, ...
11             'DisplayName', ['a = ', num2str(a)]);
12 end
```

1.2.4 Comments or explanation

The semilogy plot exposes side lobes that are invisible on a linear scale. As a approaches 0, the side lobes contain significantly more energy and decay slowly. As a increases to 1, the main lobe becomes wider, but out-of-band energy is suppressed.

1.3 A.3 Theoretical and Practical Bandwidth

1.3.1 Statement of the question

Compute the theoretical bandwidth and find the approximate practical bandwidth utilizing horizontal thresholds $c = T/10^3$ and $c = T/10^5$.

1.3.2 Comments or explanation

Theoretical Bandwidth: $BW = \frac{1+a}{2T}$. For $T = 10^{-2}$ s, BW is 50 Hz, 75 Hz, and 100 Hz for $a = 0, 0.5, 1$ respectively.

Practical Bandwidth: For $c = T/10^3$, the $a = 1$ pulse decays below the threshold quicker despite the wider main lobe, proving more bandwidth-efficient for strict constraints.

2 Part B: Orthonormality Properties

2.1 B.1 Shifts and Integrals

2.1.1 Statement of the question

Plot $\phi(t)$, $\phi(t - kT)$, and their product for $a = 0.5$ and $k = 0, 1, 2, 3$. Numerically approximate the integrals of the products for $a \in \{0, 0.5, 1\}$.

2.1.2 Figure

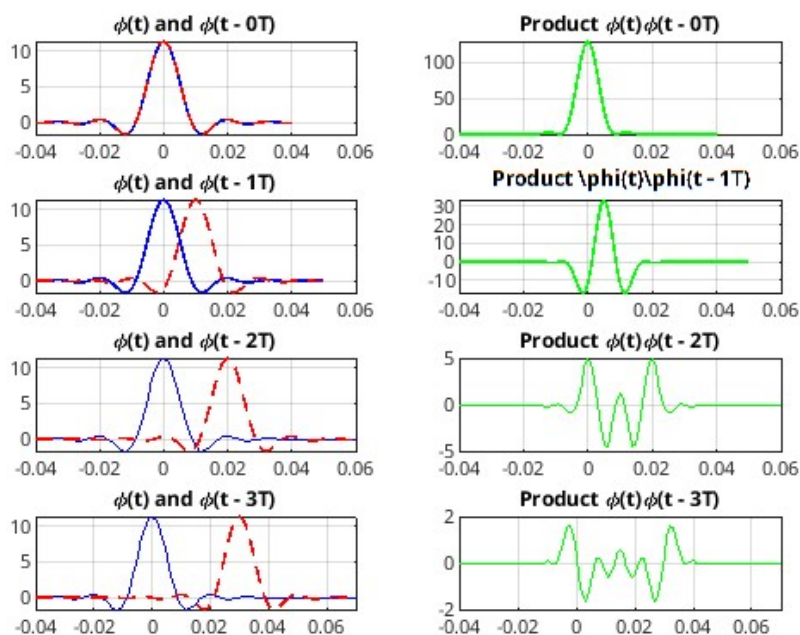


Figure 3: Pulse shifts and their products for $k = 0, 1, 2, 3$ and $a = 0.5$.

2.1.3 MATLAB Code

```

1 % Interpolate to common axis and multiply
2 t_comm = min(t_temp(1), t_shift(1)) : Ts : max(t_temp(end), t_shift(end));
3 p1 = zeros(size(t_comm)); p2 = zeros(size(t_comm));
4 s1 = find(abs(t_comm - t_temp(1)) < 1e-10);
5 p1(s1 : s1 + length(t_temp) - 1) = phi_temp;
6 s2 = find(abs(t_comm - t_shift(1)) < 1e-10);
7 p2(s2 : s2 + length(phi_temp) - 1) = phi_temp;
8 integral_val = sum(p1 .* p2) * Ts;

```

2.1.4 Comments or explanation

Truncated pulses are only approximately orthonormal. The pulse with $a = 1$ yields the best approximation because its tails contain nearly zero energy.

3 Part C: Baseband PAM System Simulation

3.1 C.1 Signal Generation and Symbol Mapping

3.1.1 Statement of the question

Generate $N = 50$ random bits b_i and write a function `bits_to_2PAM(b)` to map $0 \rightarrow +1$ and $1 \rightarrow -1$.

3.1.2 MATLAB Code

```
1 %% bits_to_2PAM.m
2 function [X] = bits_to_2PAM(b)
3 for i = 1:length(b)
4     if (b(i) == 0)
5         X(i) = 1;
6     elseif (b(i) == 1)
7         X(i) = -1;
8     end
9 end
10 end
```

3.2 C.2 System Modeling

3.2.1 Simulation of $X_\delta(t)$

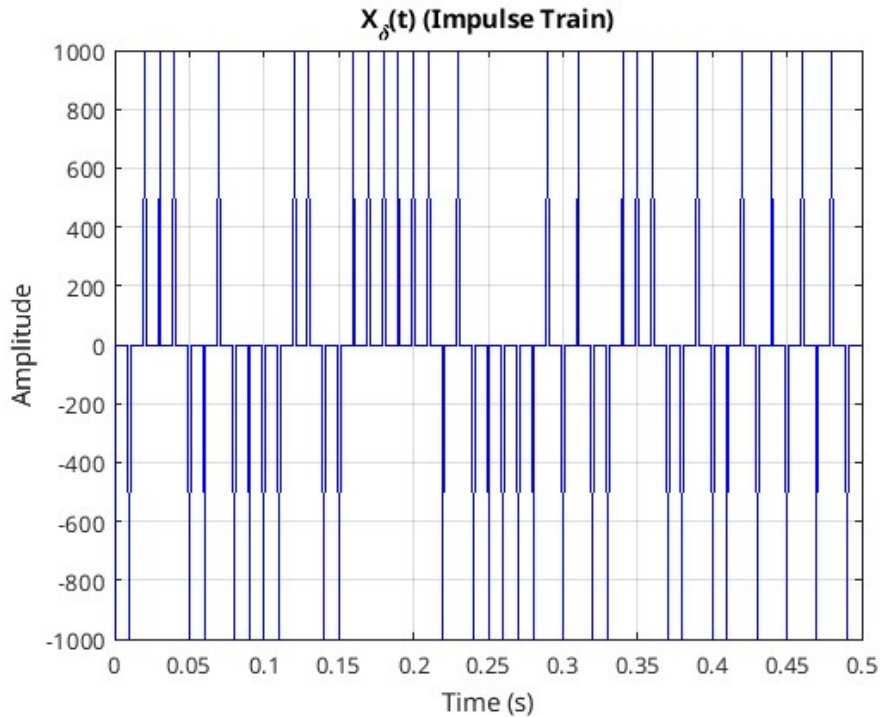


Figure 4: Impulse train $X_\delta(t)$ for the transmitted sequence.

3.2.2 Simulation of Transmitted Signal $X(t)$

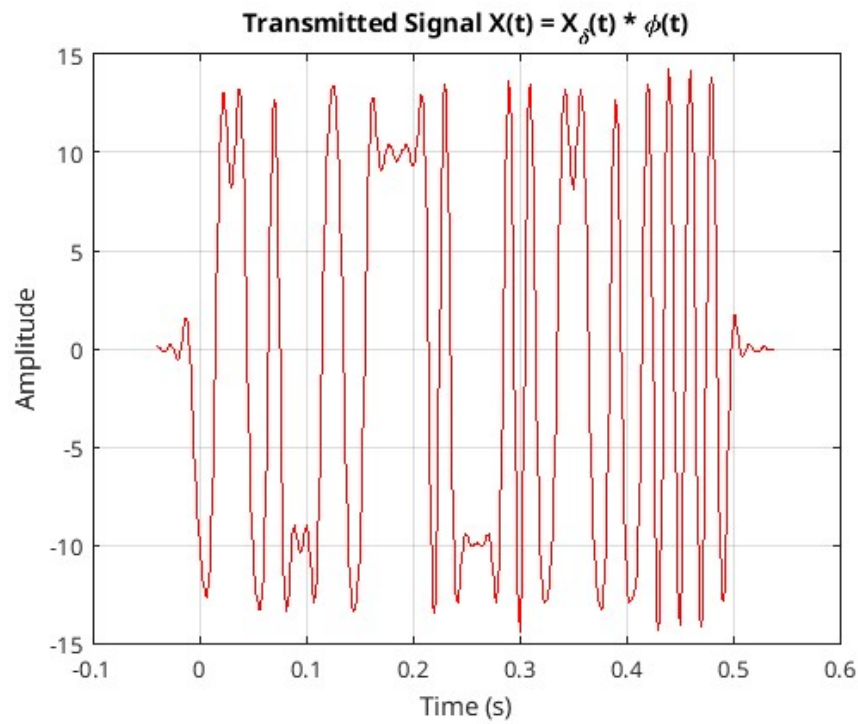


Figure 5: Transmitted baseband signal $X(t)$.

3.2.3 Receiver Matched Filter and Sampling

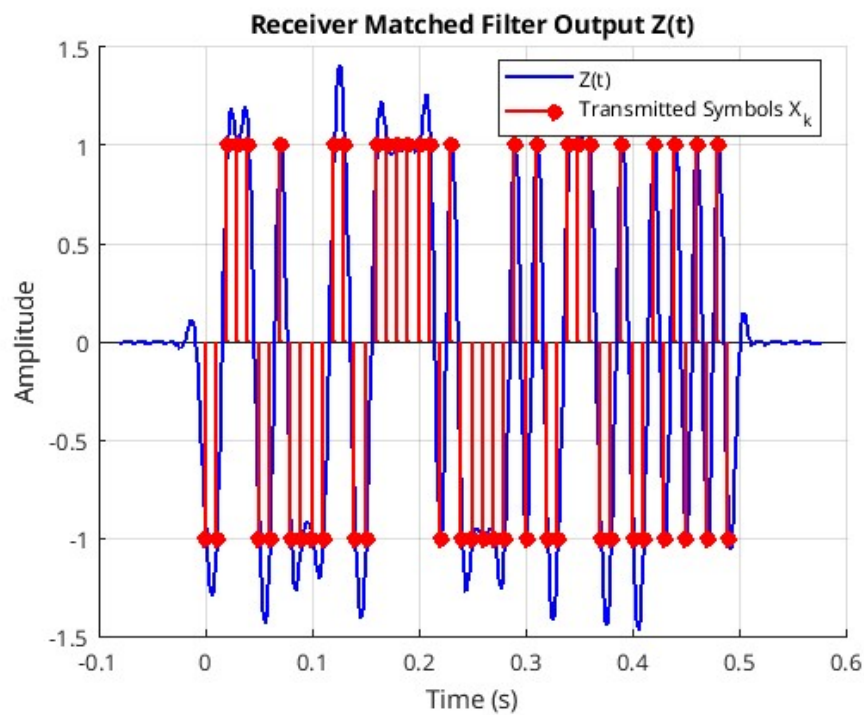


Figure 6: Matched filter output $Z(t)$ with sampled symbols X_k .

3.2.4 Comments or explanation

X_k symbols align with $Z(t)$ peaks at $t = kT$. This occurs because $\phi(t) * \phi(-t)$ forms a Raised Cosine pulse satisfying the Nyquist criterion for zero Inter-Symbol Interference (ISI).